

EDUCATION

Massachusetts Institute of Technology		
2018–2024	Ph.D. Electrical Engineering & Computer Science	Minor in Science, Technology, and Society
	Advisor: Arvind Satyanarayan	
2018–2020	S.M. Electrical Engineering & Computer Science	
Princeton University		
2014–2018	B.S.E. Computer Science	
	Advisor: J. Nathan Matias	
	Certificate in Visual Arts	
	Advisor: David Reinfurt	
University of Oxford		
Spring 2017	Visiting Student	

EMPLOYMENT

University of Colorado Boulder		
2025–	Assistant Professor	
	Department of Information Science	
	Department of Computer Science (Courtesy)	
Harvard University, Berkman Klein Center for Internet and Society		
2025–	Faculty Associate	
2024–2025	Fellow	
Massachusetts Institute of Technology, Computer Science and Artificial Intelligence Laboratory (CSAIL)		
2024–2025	Postdoctoral Associate	
2018–2024	Graduate Researcher	
Cornell University, Citizens and Technology Lab		
Summer 2020	Graduate Research Fellow	
Google, AI UX Team		
Summer 2018	UX Engineering Intern	
Google, Material UX Team		
Summer 2017	UX Engineering Intern	
Linked by Air		
Summer 2016	Software and Design Intern	
Square		
Summer 2015	Software Engineering Intern	
Coursera		
Summer 2014	Software Engineering Intern (KPCB Engineering Fellow)	

AWARDS & HONORS

2026	de Castro Award, CU Boulder CMDI
2023	Forbes 30 Under 30 (Science)
2022	MIT Morningside Academy for Design Fellowship
	MIT Prize for Open Data Honorable Mention
	Fast Company Innovation By Design Awards Finalist
2020	Schnitzer Prize in the Visual Arts, MIT
2019	Paul and Daisy Soros Fellowship for New Americans
	NSF Graduate Research Fellowship
	Hertz Foundation Fellowship Finalist

2018	Lucas Award in Visual Arts, Princeton University NSF Graduate Research Fellowship Honorable Mention
2017	Santos-Dumont Prize for Innovation, Princeton University Tau Beta Pi Scholarship Students All to Chaumont 3rd Place, Centre National du Graphisme Peter B. Lewis Summer Fund for the Arts
2016	Facebook Global Hackathon Finals Honorable Mention
2015	Greylock Partners Hackfest, 1st Place
2014	HackRice, 1st Place, Rice University KPCB Engineering Fellowship MIT THINK Scholars Interact Fellowship
2012	Eagle Scout

PUBLICATIONS

Peer-Reviewed Publications

2025. **Benthic: Perceptually Congruent Structures for Accessible Charts and Diagrams.** Catherine Mei*, Josh Pollock*, Daniel Hajas, [Jonathan Zong](#), Arvind Satyanarayan. *ACM SIGACCESS Conference on Computers and Accessibility (ACM ASSETS)*.

2025. **Tactile Vega-Lite: Rapidly Prototyping Tactile Charts with Smart Defaults.** Mengzhu (Katie) Chen, Isabella Pedraza Pineros, Arvind Satyanarayan, [Jonathan Zong](#). *ACM Conference on Human Factors in Computing Systems (ACM CHI)*.

2024. **Umwelt: Accessible Structured Editing of Multi-Modal Data Representations.** [Jonathan Zong](#), Isabella Pedraza Pineros, Mengzhu (Katie) Chen, Daniel Hajas, Arvind Satyanarayan. *ACM Conference on Human Factors in Computing Systems (ACM CHI)*.

2024. **"Customization is Key": Reconfigurable Textual Tokens for Accessible Data Visualizations.** Shuli Jones, Isabella Pedraza Pineros, Daniel Hajas, [Jonathan Zong](#), Arvind Satyanarayan. *ACM Conference on Human Factors in Computing Systems (ACM CHI)*.

2024. **Beyond Dark Patterns: A Concept-Based Framework for Ethical Software Design.** Evan Caragay, Katherine Xiong, [Jonathan Zong](#), Daniel Jackson. *ACM Conference on Human Factors in Computing Systems (ACM CHI)*.

2024. **Data Refusal From Below: A Framework for Understanding, Evaluating, and Envisioning Refusal as Design.** [Jonathan Zong](#), J. Nathan Matias. *ACM Journal on Responsible Computing*.

2023. **Animated Vega-Lite: Unifying Animation with a Grammar of Interactive Graphics.** [Jonathan Zong](#)*, Josh Pollock*, Dylan Wootton, Arvind Satyanarayan. *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS)*.

2022. **Rich Screen Reader Experiences for Accessible Data Visualization.** [Jonathan Zong](#)*, Crystal Lee*, Alan Lundgard*, JiWoong Jang, Daniel Hajas, Arvind Satyanarayan. *Computer Graphics Forum (EuroVis)*. **Best Paper Honorable Mention**

2022. **Bartleby: Procedural and Substantive Ethics in the Design of Research Ethics Systems.** [Jonathan Zong](#), J. Nathan Matias. *Social Media + Society*.

2021. **Lyra 2: Designing Interactive Visualizations by Demonstration.** [Jonathan Zong](#), Dhiraj Barnwal, Rupayan Neogy, Arvind Satyanarayan. *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS)*.

2020. **Representing Real-Time Multi-User Collaboration in Visualizations.** Rupayan Neogy, [Jonathan Zong](#), Arvind Satyanarayan. *IEEE Visualization Conference (VIS)*.

2019. **Unruly Cyborgs: The Relational Set Designs of Isamu Noguchi.** [Jonathan Zong](#). *Liminalities: A Journal of Performance Studies*.

Edited Publications

2023. **Pointing, Mutual Intelligibility, and the Seeing Subject in HCI.** [Jonathan Zong](#). *Interface Critique*.

2020. **From Individual Consent to Collective Refusal.** [Jonathan Zong](#). *ACM XRDS (Crossroads)*.

Workshops, Conferences, Posters, and Demos

2026. **Semantic Scaffolding: Augmenting Textual Structures with Domain-Specific Groupings for Accessible Data Exploration.** [Jonathan Zong](#)^{*}, Isabella Pedraza Pineros^{*}, Mengzhu (Katie) Chen, Daniel Hajas, Arvind Satyanarayan. *Symposium on AI and Cognitive Sciences in Education*.

2025. **Using Real Names of Disabled Participant-Contributors to Practice Citational Justice in Accessibility.** [Jonathan Zong](#). *IEEE VIS Accessibility Workshop*.

2022. **Olli: An Extensible Visualization Library for Screen Reader Accessibility.** Matt Blanco, [Jonathan Zong](#), Arvind Satyanarayan. *IEEE VIS Posters*.

2021. **Design Spaces of Domain-Specific Languages: Comparing and Contrasting Approaches in PL and HCI.** [Jonathan Zong](#)^{*}, Josh Pollock^{*}, Dylan Wootton, Arvind Satyanarayan. *PLATEAU Workshop on the Intersection of HCI and PL*.

2021. **Pointing, Mutual Intelligibility, and the Seeing Subject in HCI.** [Jonathan Zong](#). *Computer Mouse Conference*.

2020. **The (Minimal) Persuasive Advantage of Political Video over Text.** Chloe Wittenberg, Adam Berinsky, [Jonathan Zong](#), David Rand. *International Conference on Computational Social Science*.

2020. **Lyra: An Interface for Designing Interactive Visualizations.** [Jonathan Zong](#), Dhiraj Barnwal, Rupayan Neogy, Arvind Satyanarayan. *Computation + Journalism Symposium*.

2020. **Collective Refusal and the Limits of Informed Consent for Data Protection.** [Jonathan Zong](#), J. Nathan Matias. *Data & Society Contested Data Workshop*.

2018. **Automated Debriefing: Interface for Large-Scale Research Ethics.** [Jonathan Zong](#), J. Nathan Matias. *Companion of the 2018 ACM Conference on Computer Supported Cooperative Work and Social Computing*.
Best Demo Award

Theses

2024. **Designing for Participation and Power in Data Collection and Analysis.** *Ph.D. Thesis, Massachusetts Institute of Technology*.

2018. **Empirically Studying Research Ethics with Interface Designs for Debriefing Online Field Experiments.** *B.S.E. Thesis, Princeton University*.

TALKS & PRESENTATIONS

Invited Talks

Data Refusal in Conversation with *List Projects 31: Kite*

2025. MIT List Visual Arts Center

Accessible Data Representations for Screen Reader Users

2026. UCLA CS

2024. Fathom Information Design

2024. Quantitative and Digital Humanities Speaker Series, Emory University

Participation and Power in Data Collection and Analysis

2026. VIS Seminar, University of Utah

2025. HCI Seminar, Columbia University

2024. Computer Science Seminar, Brandeis University

2024. Information Science Seminar, University of Colorado Boulder

2024. Ethical Tech Working Group, Berkman Klein Center for Internet and Society

2023. Penn HCI Group, University of Pennsylvania

Accessible Data Visualization for Blind and Low Vision Users

2023. MIT Morningside Academy for Design

Designing Data Representations for Embodiment and Refusal

2022. Information Science Seminar, Cornell University

Rich Screen Reader Experiences for Accessible Data Visualization

- 2022. HackNY Fellowship Speaker Series
- 2022. Design for Assistive Technology, MIT Beaver Works Summer Institute
- 2022. DiverseAbilities Employee Resource Group, New York Department of Transportation

Designing Research Ethics Systems

- 2022. Accountable Algorithms Reading Group, Northeastern University

Art, Concepts, Visualizations

- 2022. Interact Fellowship Symposium

Designing Systems to Theorize Collective Power

- 2020. Digital Theory Lab, NYU Media, Culture, and Communication

Layered Abstractions for Interactive Visualization Design

- 2020. Digital Structures Group, MIT Department of Architecture

“The Most Liberal of Arts”

- 2019. Lewis Center for the Arts, Princeton University

Public Accountability in Research Ethics

- 2018. CivilServant Research Summit, MIT Media Lab

Conference Talks and Panels

Decentering Vision in Accessible Visualization

- 2025. Information Plus Conference

Accessible Data Visualization for Screen Reader Users

- 2025. Access, Innovation & Disability Symposium, CU Boulder
- 2025. Inclusive Design 24

Olli: Accessible Data Representations for Screen Reader Users

- 2023. Information+ Conference

Data Refusal From Below: A Framework for Understanding, Evaluating, and Envisioning Refusal as Design

- 2023. Panel: Resistance · AoIR 2023

Consent: A Research and Design Lens for Human-Computer Interaction (Panel)

- 2022. CSCW 2022

Public Display and Biometric Sans: Embodied Writing in the Age of Data

- 2020. Before and Beyond Typography, Stanford University Department of History

In Our Own Image: The Autonomy of Artificial Bodies in Synthetic Porn

- 2019. Panel: Internet Connection · Theorizing the Web

Invited Seminar Participation

What You Hear is What You See? Integrating Sonification and Visualization

- 2025. Schloss Dagstuhl – Leibniz Center for Informatics

PUBLIC SCHOLARSHIP

Op-Eds

2026. **The Web Is Being Made Accessible for AI, Not People.** [Jonathan Zong](#), Frank Elavsky. *Tech Policy Press*.

2019. **The Struggle to Democratize MIT.** Rodrigo Ochigame, Alan Lundgard, Alonso Espinosa Dominguez, [Jonathan Zong](#), Gabrielle Ballard, Sally Haslanger, Edmund Bertschinger, Mani Mengiste, Ruth Perry. *The Tech (MIT)*.

2019. **Consent Is Not an Ethical Rubber Stamp.** Crystal Lee*, [Jonathan Zong](#)*. *Slate*.

2018. **Researching the Researchers: Ethics and Privacy in Institutional Research.** [Jonathan Zong](#). *Nassau Weekly*.

Conversations

2022. **CAPSULE No. 1.** Cammie Lee, Megan Pai.

2021. **Form, Content, Data, Bodies: Jonathan Zong and Beth Semel on Biometric Sans.** [Jonathan Zong](#), Beth Semel. *Somatosphere*.

2021. **Holly Herndon in conversation with Tabea Nixdorff and Jonathan Zong.** *Version Space*.

2018. **w-t-f: a conversation with Eric Li and Jonathan Zong.** *Lewis Center for the Arts, Princeton University*.

Data Journalism

2021. **The Data Visualizations Behind COVID-19 Skepticism.** Crystal Lee, [Jonathan Zong](#), Anna Arpaci-Dusseau, Katherine Huang, Mateo Monterde, Ethan Nevidomsky, Tanya Yang, Anna Meurer, Soomin Chun, Arvind Satyanarayan. *MIT Visualization Group*.

Policy Engagement

2024. **Respecting blind/low-vision user agency when generating AI descriptions.** [Jonathan Zong](#), Arvind Satyanarayan. *Public comment to U.S. Access Board*.

EXHIBITIONS

2023 **Surveillance: From Vision to Data.** *Collection of Historical Scientific Instruments, Harvard University*.

2020 **WRECKED ALPHABET.** *Marcel Broodthaers Society of America*.
BODY TEXT. *Wiesner Gallery, MIT*.

2018 **w-t-f.** *Lucas Gallery, Princeton University*.
Internet Yami-Ichi. *Rhizome, New Art Dealers Alliance*.

2017 ***a graphic design exhibition*.** *Princeton University and IN-FO.CO, Los Angeles*.
Faire Signe. *Biennale de design graphique, Chaumont, France*.

COLLECTIONS

2018 **Van Alan Clark, Jr. Library, School of the Museum of Fine Arts, Tufts University**
Everyone Should

2016 **Graphic Arts Collection, Firestone Library, Princeton University**
Nap Poems

ART BOOK FAIRS

2019 **Tiny Tech Zines, Navel LA**
with Free Radicals
Press Play, Pioneer Works
with Tech Zine Fair

2018 **Odds and Ends, Yale University**
Zine Soup 3.0: Makers Market, Chinatown Soup
NY Tech Zine Fair, School for Poetic Computation
Boston Art Book Fair, Boston Center for the Arts

TEACHING

Instructor

2026. **Design for Accessibility.** University of Colorado Boulder

2025. **User-Centered Design.** University of Colorado Boulder

2020. **AI, Justice, Imagination: Technology and Speculative Fiction.** Tufts University Experimental College
Visiting Lecturer (Adjunct) (co-instructor with Janet Zong York)

Guest Lecturer / Speaker

2025. **Intro to Sociotechnical Systems and HCI.** Brown University

2023. **Crowd Sourced City: Data Activism Edition.** MIT Department of Urban Studies and Planning

2023. **Transversal Design for Social Impact.** MIT Program in Art, Culture, and Technology

2022. **Transformations in Engineering and the Arts.** Princeton University

Guest Critic

2022. **i-n-t-e-r-f-a-c-e**. Princeton University

MENTORING & ADVISING**M.S. Advisees**

Matt Nesselrodt

Ph.D. Dissertation Committees

Everley Tseng (CS)

Ph.D. Preliminary Exam & Proposal Committees

Blakeley H. Payne

Everley Tseng (CS)

M.Eng. Students (mentored as a Ph.D. student)

2023–24 Isabella Pedraza Pineros, Katie Chen

Summer 2023 Evan Caragay (**Charles & Jennifer Johnson MEng Thesis Award**), Katherine Xiong

2022–23 Shuli Jones (**David Adler MEng Thesis Award**)

2020–21 EJ Sefah, Katie Bacher

2019–20 Rupayan Neogy

Undergraduate Students (mentored as a Ph.D. student)

Spring 2023 Isabella Pedraza Pineros

Spring/Summer 2022 Matt Blanco (Northeastern)

Summer 2021 JiWoong Jang (CMU)

Fall 2020 Anna Arpaci-Dusseau, Katherine Huang, Mateo Monterde, Ethan Nevidomsky, Tanya Yang, Anna Meurer, Soomin Chun

SERVICE**University Service**

2025–2026 Graduate Committee, CU Boulder Department of Information Science

Peer Reviewing

2026 IEEE VIS, ACM ASSETS

2025 ACM CHI, Eurovis, IEEE TVCG, IEEE VIS, Journal of Visualization and Interaction

2024 ACM CHI, Research Ethics (SAGE Journal), ACM DIS, Data by Design (community review for MIT Press book), Journal of Visualization and Interaction, Assistive Technology (T&F Journal)

2023 ACM CHI, IEEE VIS, IEEE TVCG, ACM SIGGRAPH, PacificVis Visual Data Storytelling Contest

2022 ACM CHI (**Special Recognition for Outstanding Review**), IEEE VIS, Journal of Empirical Research on Human Research Ethics

2021 IEEE VIS

Other Academic Service

2024 Selection Committee, MIT Student Art Awards

Event Organizing

2022 Princeton University Workshop on Ethics of Field Experiments

Community Building & Activism

2020–2022 MIT Graduate Student Union, Organizer

2020–2021	MIT Human-Computer Interaction, Social Committee
2019–20	MIT Students Against War

SELECTED PRESS

Featured

2025. **A new way to make graphs more accessible to blind and low-vision readers.** *MIT News*.
2024. **Military Eye Injuries With Risk of Cataracts and Umwelt Software For Accessible Charts.** *WYPL-FM 89.3 Eye on Vision*.
2024. **New software enables blind and low-vision users to create interactive, accessible charts.** *MIT News*.
2024. **Q&A: How refusal can be an act of design.** *MIT News*.
2023. **Improving accessibility of online graphics for blind users.** *MIT News*.
2023. **Student Spotlight: Jonathan Zong.** *MIT CSAIL*.
2023. **MIT MAD Fellow Improves Accessibility of Online Graphics for Blind Users.** *MIT Morningside Academy for Design*.
2023. **These tools help visually impaired scientists read data and journals.** *Nature Technology Features*.
2022. **Developing community around design.** *MIT News*.
2022. **Making data visualization more accessible for blind and low-vision individuals.** *MIT News*.
2022. **3 Questions: Designing software for research ethics.** *MIT News*.
2021. **Helping data journalists create interactive graphics.** *MIT CSAIL News*.
2020. **Forty artists elide, dismantle, and politicize the alphabet.** *Art & Education*.
2020. **Students Plumb Barriers of Sound, Space, and Sight at MIT.** *Arts at MIT*.
2020. **Bezos, Bodies, and Backprop.** *The Tech (MIT)*.
2018. **PolitEcho: A Mirror for the “Daily Me”.** *Frontier Magazine*.
2017. **Challenge everything you think — democracy depends on it.** *The Guardian*.
2017. **How to Escape Your Political Bubble for a Clearer View.** *The New York Times*.
2016. **This browser extension shows how biased your social media feed is.** *A.V. Club*.
2016. **Find out just how biased your Facebook feed is with this Chrome extension.** *Mashable*.
2016. **A Browser Extension That Shows You Your Filter Bubble.** *New York Magazine*.
2016. **Transformations: Students find creativity at intersection of art and engineering.** *Princeton University News*.
2013. **Houston’s Hackers Gather to Help City Solve Problems.** *Xconomy*.

Quoted

2023. **Making chemistry accessible to all.** *Dell Technologies Perspectives*.
2022. **MIT grad students vote to form labor union.** *Boston Business Journal*.
2022. **After weeks of wrangling, MIT’s grad student union will come to a vote this week.** *WBUR*.
2021. **Digital exclusions: mental health and digital life.** *Our Sick Society Podcast*.
2020. **Facial Recognition Software: Masks, Ethics & The Future.** *Forbes*.
2019. **The Dark Implications of Facial Swap Filter Technology.** *Paper Magazine*.